SBE16plusV2 SeaCAT Moored

Instrument Configuration

Instrument Serial Number:	16-50676
Instrument Firmware Version:	3.2.1
Zero Conductivity Frequency:	2669.27
Communications Format:	RS232
Communications Settings:	9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

Data Format	*Measurement*	*Sensor Type*	*Serial Number*	*Rating*
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	12464170	600m(600 dBar)

Maximum Depth: 600m

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.

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SENSOR SERIAL NUMBER: 50676
CALIBRATION DATE: 25-Dec-25

SBE 16plus V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

```
a0 =   1.265938e-03
a1 =   2.692084e-04
a2 =  -3.886775e-07
a3 =   1.469109e-07
```

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	562457.600	0.9999	-0.0001
4.5001	496549.000	4.5003	0.0002
15.0000	335553.700	15.0000	-0.0000
18.5000	292950.700	18.4997	-0.0003
24.0000	235530.000	24.0002	0.0002
29.0000	192156.400	29.0002	0.0002
32.5000	166100.300	32.4998	-0.0002

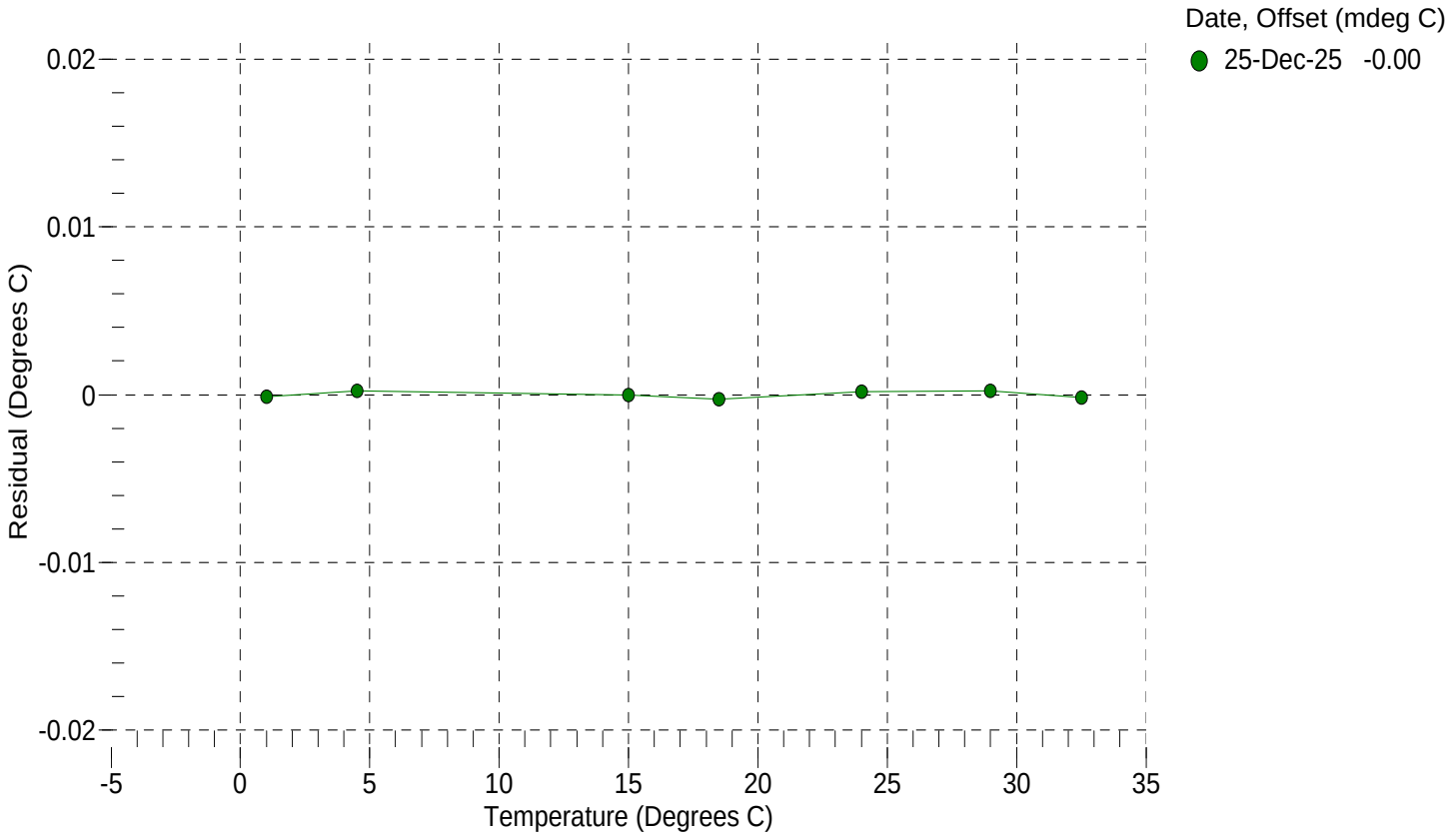
n = Instrument Output (counts)

MV = (n - 524288) / 1.6e+007

R = (MV * 2.900e+009 + 1.024e+008) / (2.048e+004 - MV * 2.0e+005)

Temperature ITS-90 (°C) = $1/\{a0 + a1[ln(R)] + a2[ln^2(R)] + a3[ln^3(R)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature



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SENSOR SERIAL NUMBER: 50676
CALIBRATION DATE: 25-Dec-25

SBE 16plus V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g =  -9.903540e-01
h =   1.392797e-01
i =  -2.026556e-04
j =   3.628963e-05

CPcor = -9.5700e-008
CTcor =  3.2500e-006
```

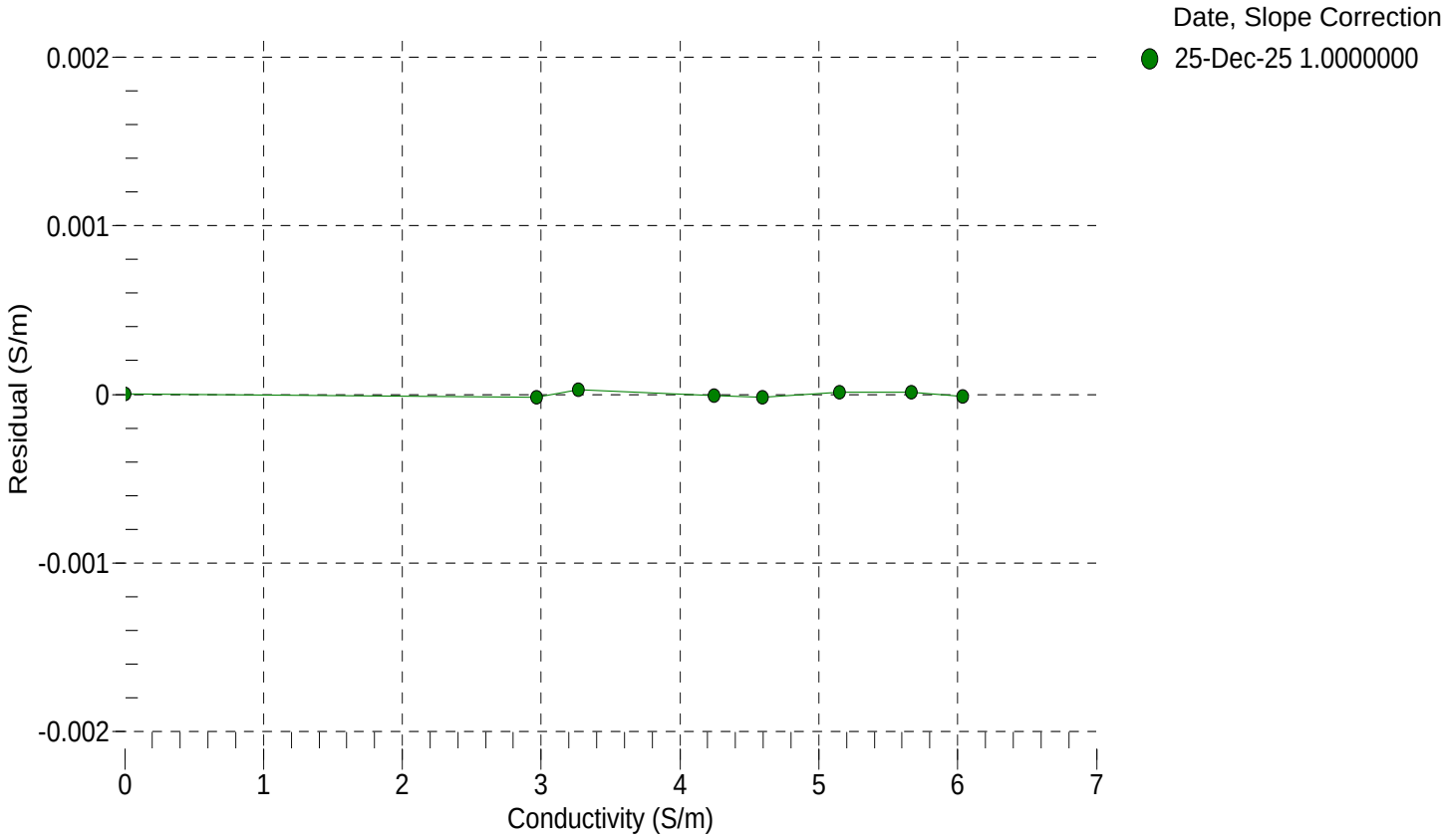
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2669.27	0.0000	0.00000
1.0000	34.6600	2.96387	5329.22	2.9639	-0.00002
4.5001	34.6398	3.26972	5530.76	3.2697	0.00003
15.0000	34.5966	4.24750	6129.88	4.2475	-0.00001
18.5000	34.5873	4.59126	6326.74	4.5912	-0.00002
24.0000	34.5770	5.14695	6632.34	5.1470	0.00001
29.0000	34.5704	5.66656	6905.49	5.6666	0.00001
32.5000	34.5657	6.03721	7093.70	6.0372	-0.00001

f = Instrument Output (Hz) / 1000.0

t = temperature (°C); p = pressure (decibars); δ = CTcor; ε = CPcor;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 50676
CALIBRATION DATE: 17-Dec-25

SBE 16plus V2 PRESSURE CALIBRATION DATA
870 psia S/N 12464170

COEFFICIENTS:

PA0 =	2.487836e+00	PTCA0 =	5.245322e+05
PA1 =	2.616607e-03	PTCA1 =	6.432549e+01
PA2 =	1.639214e-11	PTCA2 =	-9.428125e-01
PTEMPA0 =	-6.030824e+01	PTCB0 =	2.536513e+01
PTEMPA1 =	5.407550e+01	PTCB1 =	2.255639e-04
PTEMPA2 =	-3.449528e-01	PTCB2 =	0.000000e+00

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.47	530064.0	1.5	14.47	-0.00
179.74	593201.0	1.5	179.71	-0.00
359.75	661915.0	1.5	359.70	-0.01
539.76	730586.0	1.5	539.73	-0.00
719.77	799193.0	1.5	719.75	-0.00
869.79	856326.0	1.5	869.78	-0.00
719.77	799217.0	1.5	719.81	0.00
539.78	730620.0	1.5	539.82	0.00
359.78	661953.0	1.5	359.79	0.00
179.77	593240.0	1.5	179.81	0.00
14.48	530081.0	1.5	14.49	0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	1.74	530273.20
29.00	1.67	530241.20
24.00	1.57	530171.20
18.50	1.47	530048.30
15.00	1.41	529931.40
4.50	1.21	529431.80
1.00	1.14	529245.90

TEMPERATURE (°C)	SPAN
-5.00	25.36
34.90	25.37

y = thermistor output (counts)

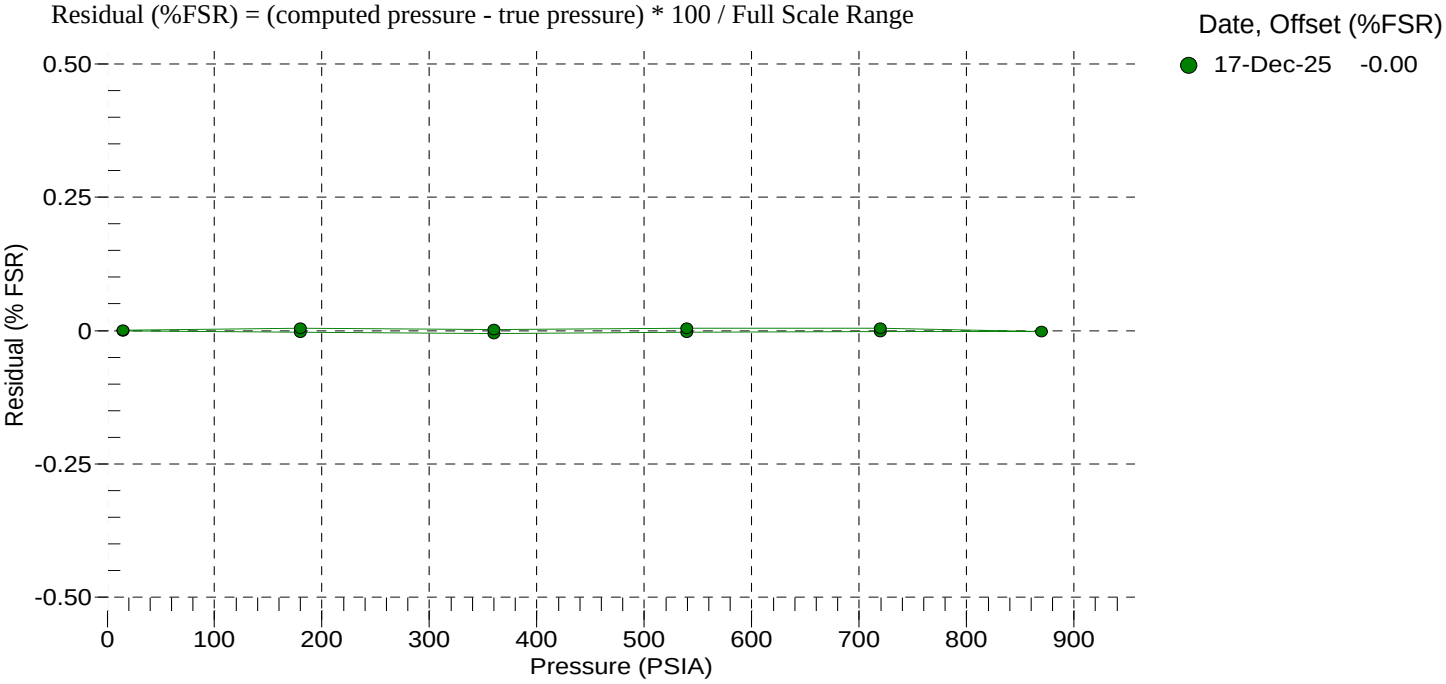
$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



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Pressure Test Certificate

Test Date: 2025-12-16

Description: SBE-16P SeaCat

Sensor Information:

Model Number: SBE-16P

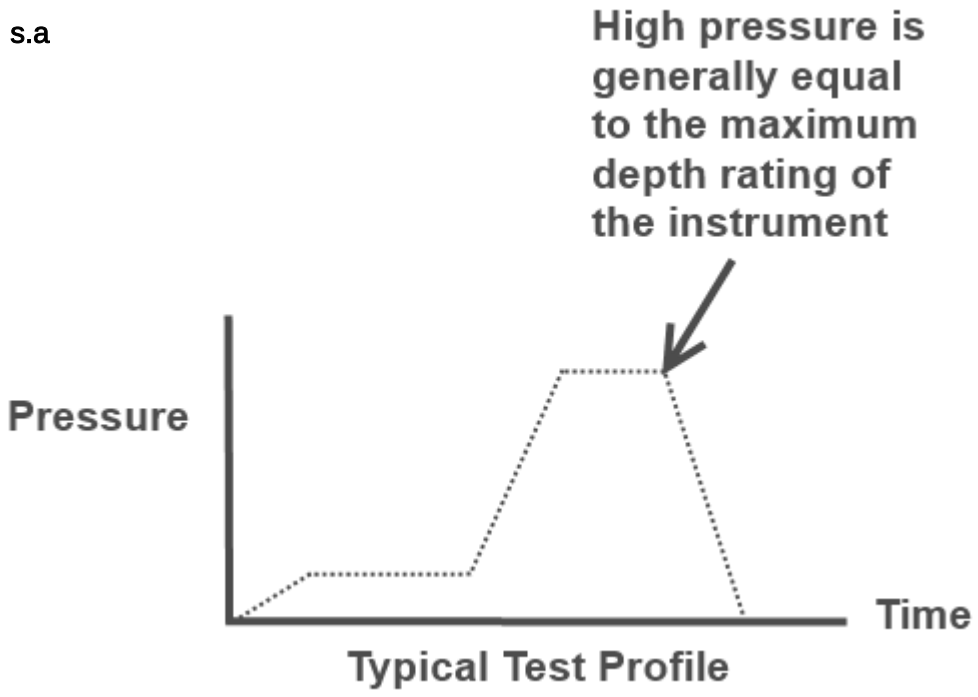
Serial Number: 50676

Pressure Test Protocol:

Low Pressure Test:	40	PSI	Held For:	15	Minutes
High Pressure Test:	870	PSI	Held For:	15	Minutes

Passed Test: True

Tested By: s.a



Typical Test Profile